

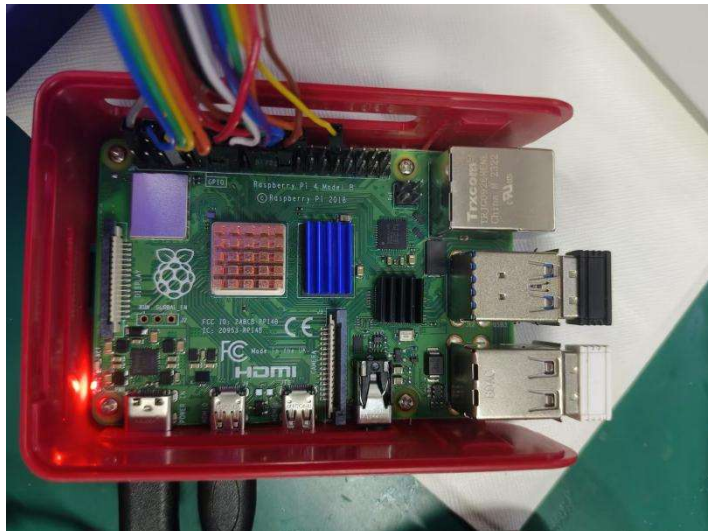
Driven LED Screen Based on Raspberry Pie

Overview:

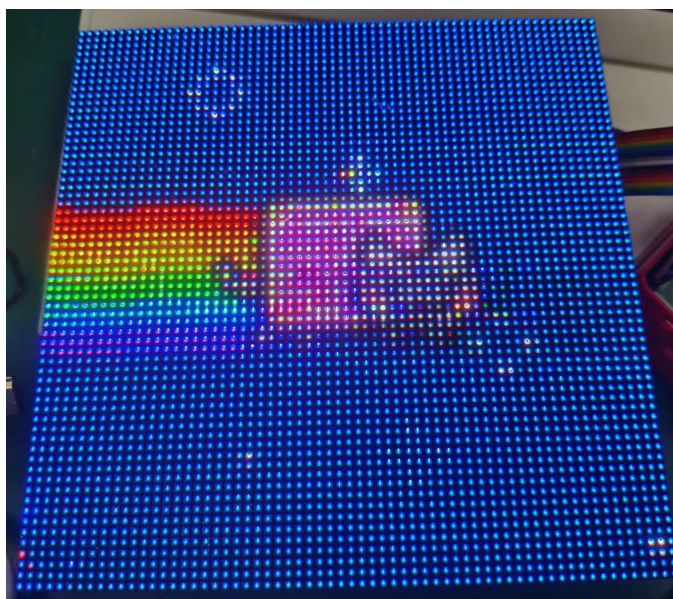
This technical document provides a comprehensive and in-depth description of how to start from scratch, combine a raspberry pie with an LED screen, successfully light up the LED screen drive, and conduct a series of creative and fun experiments.

一、Device Preparation

1、Raspberry Pie 4B



2、64*64 LED Screen



<https://github.com/derwinluan/embedded-iot-portfolio>

3、HDMI External Display

4、Mouse & Keyboard

二、Preparations

1、Hardware wiring



Just Connect Lines with Smiley Faces !

2、Raspberry Pi System Burning

Enter: <https://www.raspberrypi.com/software/>

<https://github.com/derwinluan/embedded-iot-portfolio>

Raspberry Pi OS

Your Raspberry Pi needs an operating system to work. This is it. Raspberry Pi OS (previously called Raspbian) is our official supported operating system.



Install Raspberry Pi OS using Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install Raspberry Pi OS and other operating systems to a microSD card, ready to use with your Raspberry Pi.

Download and install Raspberry Pi Imager to a computer with an SD card reader. Put the SD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.

[Download for Windows](#)

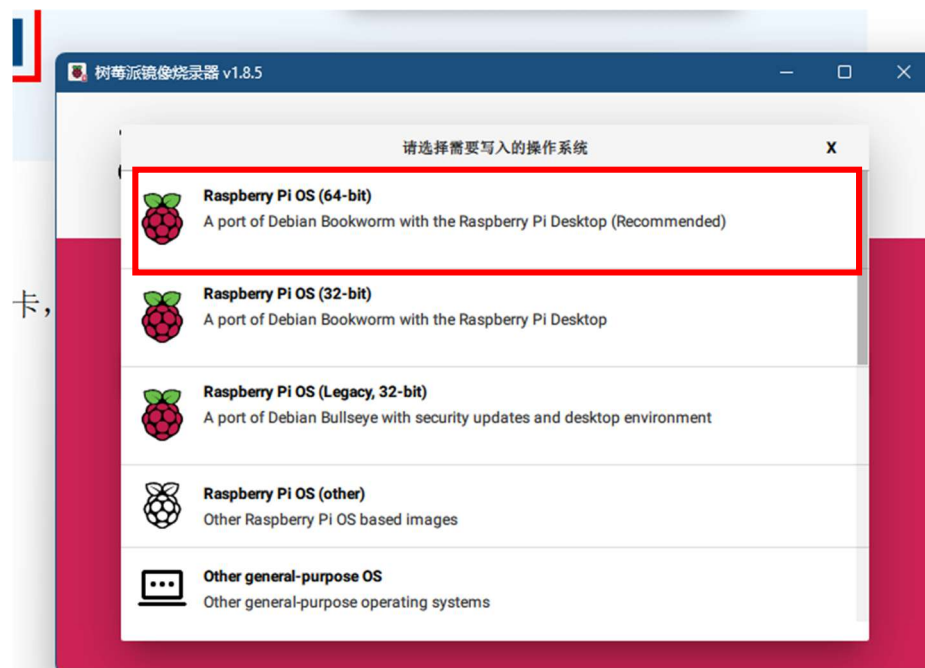
[Download for macOS](#)

[Download for Ubuntu for x86](#)

To install on **Raspberry Pi OS**, type
`sudo apt install rpi-imager`
in a Terminal window.



After the download and installation are completed, insert the memory card, select the 64-bit system, and click "Burn".



3、Power on and start up, Configure the username and password (Current password: 123456), connect WIFI。

4、Connect the display screen using an HDMI cable



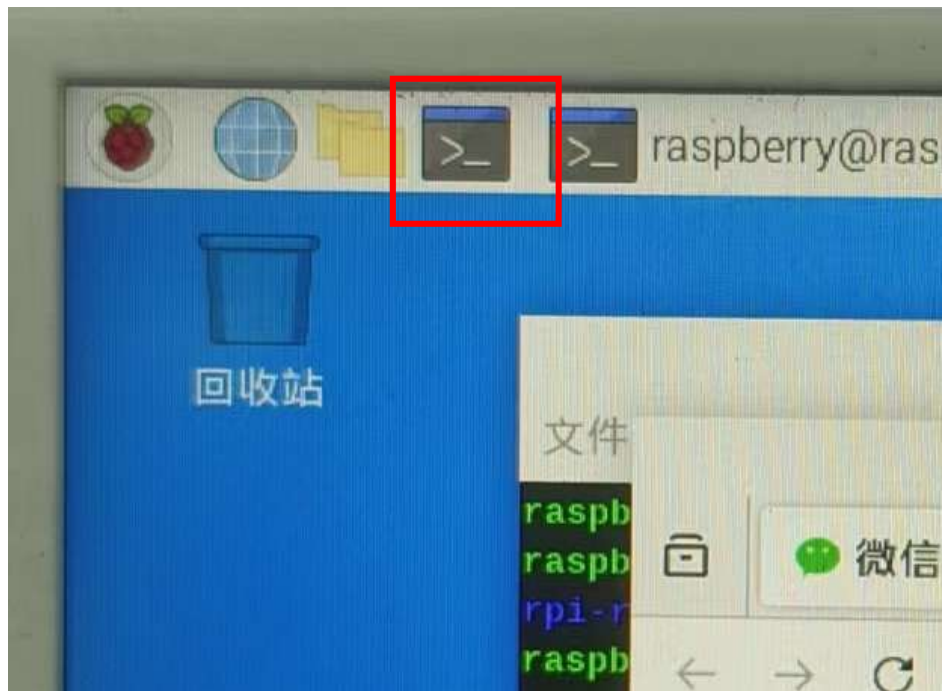
三、 Specific Steps

1、 Open the command line, create a new folder, then change the directory to that folder, and enter the command

```
"git clone https://github.com/hzeller/rpi-rgb-led-matrix.git"
```

to download the project to your local machine.

If it is a compressed file, you will need to extract it. (*Current project directory: ./home/project/rpi-rgb-led-matrix*)



```
raspberrypi@raspberrypi:~/project $ ls
raspberrypi@raspberrypi:~/project $ git clone https://github.com/hzeller/rpi-rgb-led-matrix.git
正在克隆到 'rpi-rgb-led-matrix'...
remote: Enumerating objects: 5392, done.
remote: Counting objects: 100% (30/30), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 5392 (delta 21), reused 12 (delta 12), pack-reused 5362 (from 2)
接收对象中: 100% (5392/5392), 22.39 MiB | 185.00 KiB/s, 完成。
处理 delta 中: 100% (3709/3709), 完成。
```

2、Disable the onboard audio driver of the Raspberry Pi and modify the cmdline.txt file

(1) Disable the driver

Open the command line and enter:

```
cat <<EOF | sudo tee /etc/modprobe.d/blacklist-rgb-matrix.conf
blacklist snd_bcm2835
EOF
```

Then input `sudo update-initramfs -u` upload initramfs

Finally input `sudo reboot` to restart

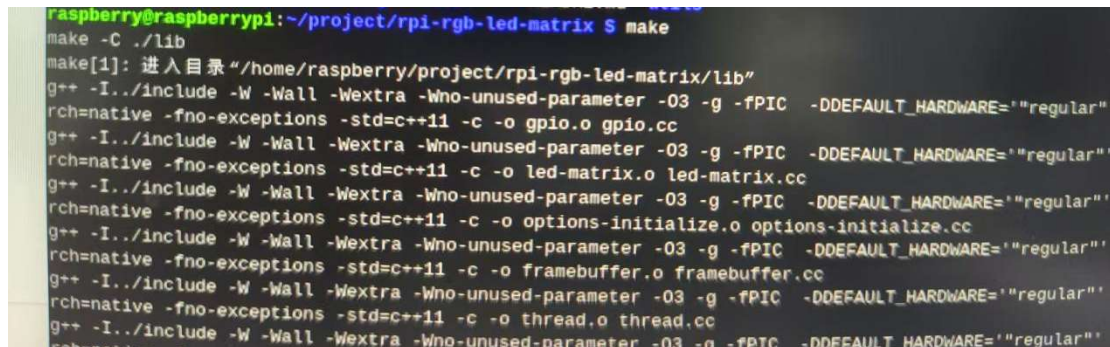
(2) Modify the file

Enter: `sudo nano /boot/cmdline.txt`

After entering the file, add "`isolcpus=3`" at the end of the line followed by a space.

Input: `CTRL+O`, press Enter, then `CTRL+X`.

3、 Enter the directory "`/home/project/rpi-rgb-led-matrix`", then type "`make`" to compile the project.

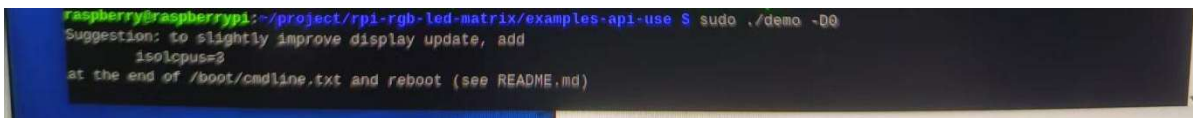


```
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix $ make
make -C ./lib
make[1]: 进入目录 "/home/raspberrypi/project/rpi-rgb-led-matrix/lib"
g++ -I../include -W -Wall -Wextra -Wno-unused-parameter -O3 -g -fPIC -DDEFAULT_HARDWARE="regular" -fcommon -fno-exceptions -std=c++11 -c -o gpio.o gpio.cc
g++ -I../include -W -Wall -Wextra -Wno-unused-parameter -O3 -g -fPIC -DDEFAULT_HARDWARE="regular" -fcommon -fno-exceptions -std=c++11 -c -o led-matrix.o led-matrix.cc
g++ -I../include -W -Wall -Wextra -Wno-unused-parameter -O3 -g -fPIC -DDEFAULT_HARDWARE="regular" -fcommon -fno-exceptions -std=c++11 -c -o options-initialize.o options-initialize.cc
g++ -I../include -W -Wall -Wextra -Wno-unused-parameter -O3 -g -fPIC -DDEFAULT_HARDWARE="regular" -fcommon -fno-exceptions -std=c++11 -c -o framebuffer.o framebuffer.cc
g++ -I../include -W -Wall -Wextra -Wno-unused-parameter -O3 -g -fPIC -DDEFAULT_HARDWARE="regular" -fcommon -fno-exceptions -std=c++11 -c -o thread.o thread.cc
g++ -I../include -W -Wall -Wextra -Wno-unused-parameter -O3 -g -fPIC -DDEFAULT_HARDWARE="regular" -fcommon -fno-exceptions -std=c++11 -c -o thread.o thread.cc
```

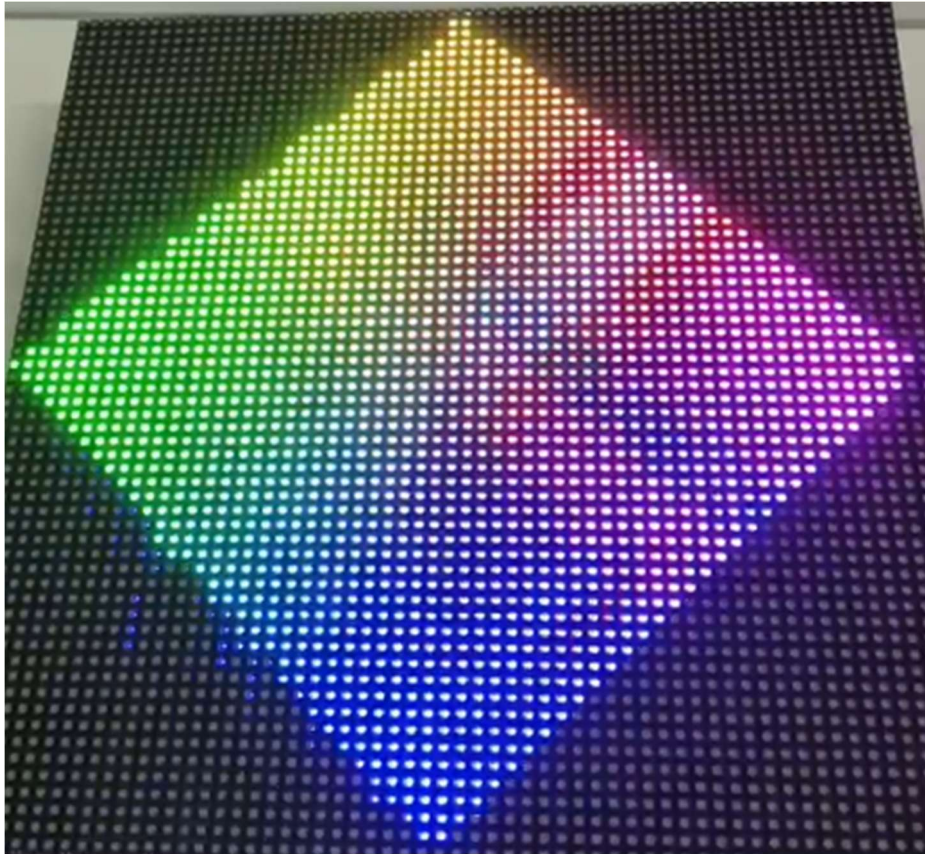
5、 After the compilation is completed, go to the directory:

`/home/project/rpi-rgb-led-matrix/examples-api-use`, and enter the command:

`sudo ./demo -D0 --led-rows=64 --led-cols=64`. The following effect will appear:



```
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix/examples-api-use $ sudo ./demo -D0
Suggestion: to slightly improve display update, add
isolcpus=3
at the end of /boot/cmdline.txt and reboot (see README.md)
```



If an error message appears:

```
=== snd_bcm2835: found that the Pi sound module is loaded. ===
```

Don't use the built-in sound of the Pi together with this lib; it is known to be incompatible and cause trouble and hangs (you can still use external USB sound adapters).

See Troubleshooting section in README how to disable the sound module.

You can also run with `--led-no-hardware-pulse` to avoid the incompatibility, but you will have more flicker.

Exiting; fix the above first or use `--led-no-hardware-pulse`

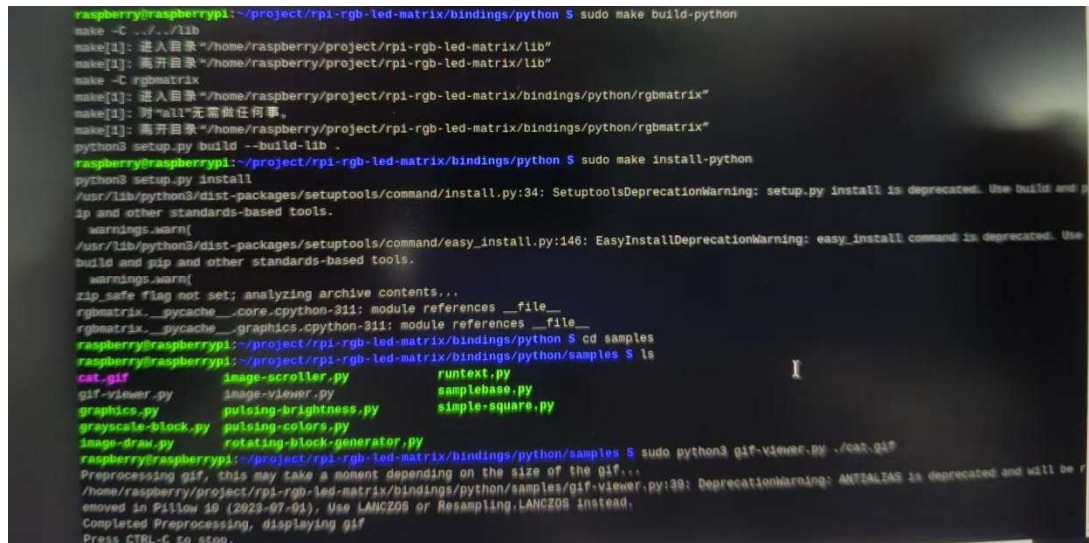
Then, the second step needs to be repeated: disable the onboard audio driver of the Raspberry Pi, and after restarting, input `"lsmod | grep snd_bcm2835"` to verify if the driver has been loaded. If it has not been loaded, it is OK.

6、Configuring the Python environment:

In the directory `./home/project/rpi-rgb-led-matrix/bindings/python`, enter the following commands: `sudo apt-get update && sudo apt-get install python3-dev cython3 -y`

Then enter: `sudo make build-python`

Finally enter: `sudo make install-python`



```
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix/bindings/python $ sudo make build-python
make -C ../lib
make[1]: 进入目录 "/home/raspberrypi/project/rpi-rgb-led-matrix/lib"
make[1]: 离开目录 "/home/raspberrypi/project/rpi-rgb-led-matrix/lib"
make -C rgbmatrix
make[1]: 进入目录 "/home/raspberrypi/project/rpi-rgb-led-matrix/bindings/python/rgbmatrix"
make[1]: 对"all"无需做任件事。
make[1]: 离开目录 "/home/raspberrypi/project/rpi-rgb-led-matrix/bindings/python/rgbmatrix"
python3 setup.py build --build-lib .
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix/bindings/python $ sudo make install-python
python3 setup.py install
/usr/lib/python3/dist-packages/setuptools/command/install.py:34: SetuptoolsDeprecationWarning: setup.py install is deprecated. Use build and pip
and other standards-based tools.
  warnings.warn(
/usr/lib/python3/dist-packages/setuptools/command/easy_install.py:146: EasyInstallDeprecationWarning: easy_install command is deprecated. Use
build and pip and other standards-based tools.
  warnings.warn(
zip_safe flag not set; analyzing archive contents...
rgbmatrix.__pycache__.core.cpython-311: module references __file__
rgbmatrix.__pycache__.graphics.cpython-311: module references __file__
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix/bindings/python $ cd samples
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix/bindings/python/samples $ ls
cat.gif          image-scroller.py      runtext.py
gif-viewer.py    image-viewer.py        samplebase.py
graphics.py      pulsing-brightness.py  simple-square.py
grayscale-block.py pulsing-colors.py
image-draw.py    rotating-block-generator.py
raspberrypi@raspberrypi:~/project/rpi-rgb-led-matrix/bindings/python/samples $ sudo python3 gif-viewer.py ./cat.gif
Preprocessing gif, this may take a moment depending on the size of the gif...
/home/raspberrypi/project/rpi-rgb-led-matrix/bindings/python/samples/gif-viewer.py:39: DeprecationWarning: ANTIALIAS is deprecated and will be r
emoved in Pillow 10 (2023-07-01). Use LANCZOS or Resampling.LANCZOS instead.
Completed Preprocessing, displaying gif
Press CTRL-C to stop.
```

7、Enter the directory: `/home/project/rpi-rgb-led-matrix/bindings/python/samples/` and modify the file `gif-viewer.py`

Input: `sudo nano gif-viewer.py` Modify the screen resolution to `64*64`


```
try:
    num_frames = gif.n_frames
except Exception:
    sys.exit("provided image is not a gif")

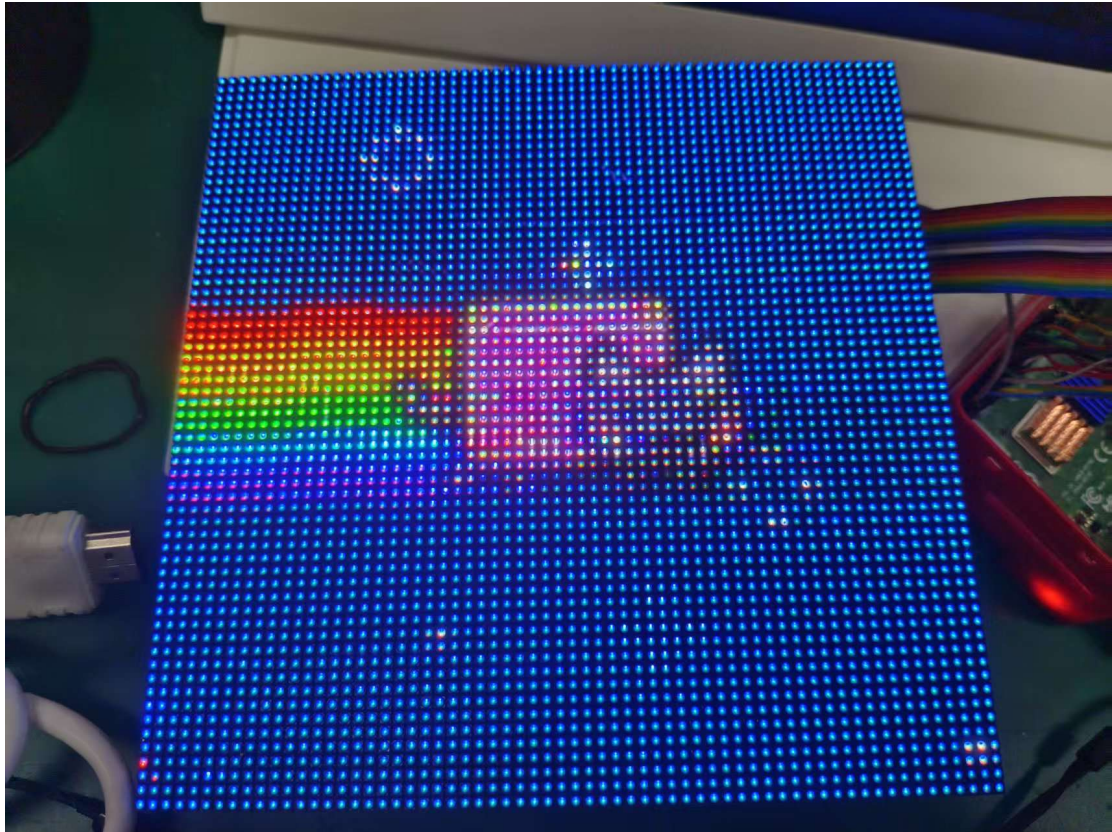
# Configuration for the matrix
options = RGBMatrixOptions()
options.rows = 32
options.cols = 32
options.chain_length = 1
options.parallel = 1
options.hardware_mapping = 'regular' # If you have an Adafruit HAT: 'adafruit-hat'

matrix = RGBMatrix(options = options)

# Preprocess the gifs frames into canvases to improve playback performance
canvases = []
print("Preprocessing gif, this may take a moment depending on the size of the gif...")
for frame_index in range(0, num_frames):
    gif.seek(frame_index)
    # must copy the frame out of the gif, since thumbnail() modifies the image in-place
    frame = gif.copy()
    frame.thumbnail((matrix.width, matrix.height), Image.ANTIALIAS)
    canvas = matrix.CreateFrameCanvas()
    canvas.SetImage(frame.convert("RGB"))
    canvases.append(canvas)
# Close the gif file to save memory now that we have copied out all of the frames
```

8、 Enter the command "*sudo python3 gif-viewer.py ./cat.gif*"(assuming there is a "cat.gif" file in the directory, which is currently included).

This will produce the following result:



To achieve the effects of other GIF animations, simply go online to download them.